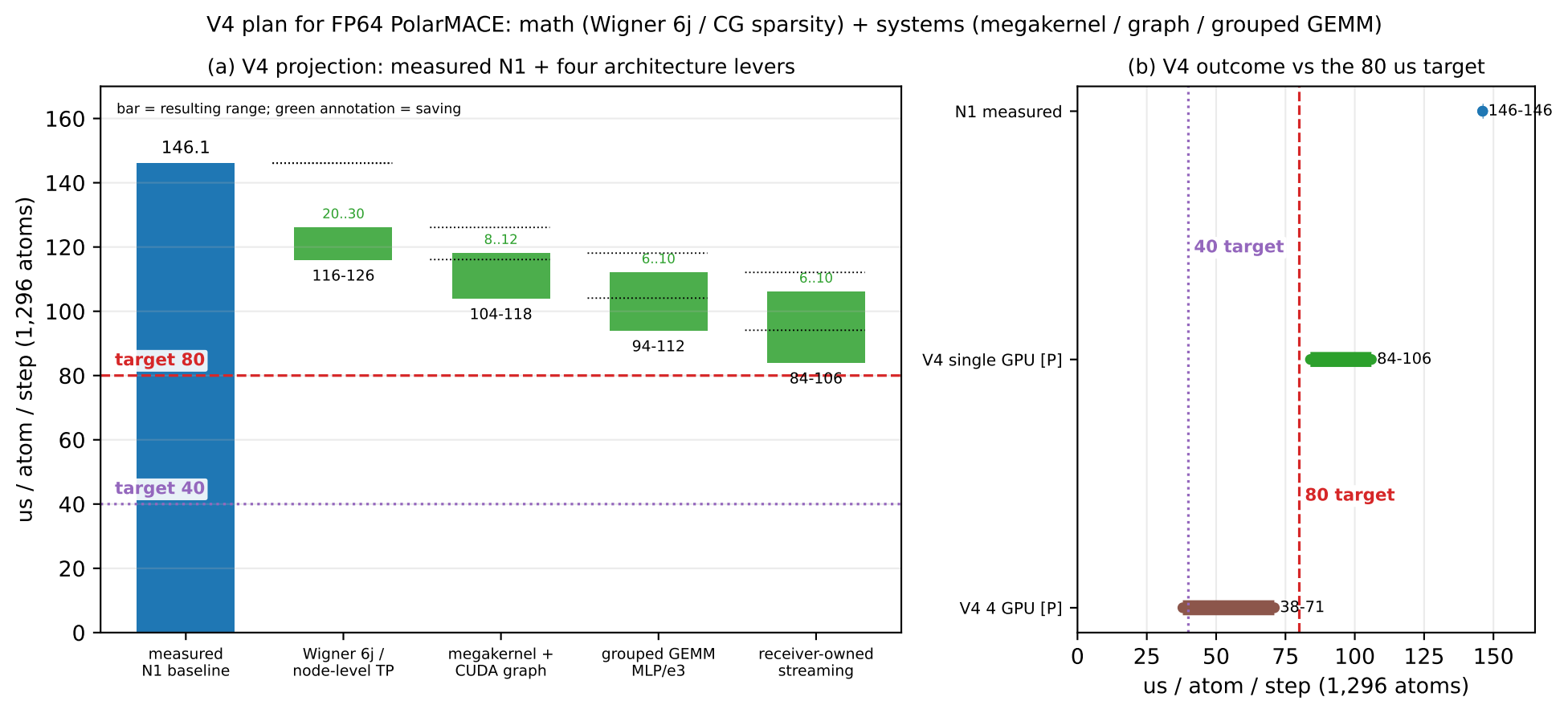
V4 plan for FP64 PolarMACE: math (Wigner 6j / CG sparsity) + systems (megakernel / graph / grouped GEMM)
(a) V4 projection: measured N1 + four architecture levers
bar = resulting range; green annotation = saving
us / atom / step (1,296 atoms)
146.1
20..30
116-126
8..12
104-118
6..10
94-112
6..10
84-106
target 80
target 40
measured N1 baseline
Wigner 6j / node-level TP
megakernel + CUDA graph
grouped GEMM MLP/e3
receiver-owned streaming
(b) V4 outcome vs the 80 us target
N1 measured
V4 single GPU [P]
V4 4 GPU [P]
146-146
84-106
38-71
40 target
80 target
us / atom / step (1,296 atoms)